



The Breast Appreciation Scale: An examination of its psychometric properties in French-Canadian women

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ARTICLE INFO

Keywords:

Breast appreciation
French
Canada
Psychometrics
Positive body image
Test adaptation

ABSTRACT

Scholars are increasingly focused on women's positive, embodied "breasted experienced", which includes breast appreciation. This positive body image facet is assessed using the 9-item Breast Appreciation Scale (BrAS; Swami et al., 2022), but assessments of its psychometric properties in diverse cultural communities are limited. Here, we report on the translation and validation of a French translation of the BrAS in French-Canadian women. A sample of 365 women from Quebec, Canada, completed the French BrAS alongside additional measures. Exploratory and confirmatory factor analysis in two split-half subsamples identified a stable unidimensional model of the BrAS, with six of the nine original items retained. This BrAS model was fully invariant across both split-half subsamples and showed no evidence of differential item functioning in relation to participant age. Latent breast appreciation also evidenced strong composite reliability, as well as adequate convergent and concurrent validity. Specifically, greater breast appreciation was associated with higher overall body appreciation, lower breast size dissatisfaction, better psychological well-being and lower distress, and lower disordered eating patterns. While the present results provide initial psychometric support for the BrAS in French-Canadian women, more work can be done to confirm and extend our findings.

1. Introduction

The breasts have historically mattered in popular and symbolic language as signifiers of a woman's sexuality or maternity (Bartlett, 2000; Lee, 1997; Young, 1992). In particular, heteronormative and patriarchal discourse frequently commodifies, objectifies, and fetishes the breasts (e.g., Gerald & Potvin, 2009; Rodriguez & Hernandez, 2019; Yockey et al., 2019). Unsurprisingly, then, a majority of women in diverse cultural settings are dissatisfied with their breasts (e.g., Fredrick et al., 2008; Lombardo et al., 2019; Swami et al., 2015). For example, one cross-national study of women from 40 nations reported that most (70%) were dissatisfied with the size of their breasts, with nearly half desiring larger breasts than they currently had (Swami et al., 2020). Breast dissatisfaction, in turn, is associated with a range of negative outcomes, including more negative body image and higher risk

of disordered eating (e.g., Swami et al., 2015), poorer psychological well-being (e.g., Koff & Benavage, 1998), poorer breast cancer awareness (Swami & Furnham, 2018), and lower sexual satisfaction (Ålgars et al., 2011).

While a focus on negative experiences related to the breasts remains important, it also risks missing the complexities of women's "breasted experiences" (Young, 1992, p. 215). For example, in a seminal article, Young (1992) described how women's sensory experiences of the breasts – such as how they move or how sensitive they are – matter more to women than the aesthetics of their breasts in the absence of a male gaze. As such, Young (1992) advocated for a greater understanding of women's positive "breasted experiences", which might include the breasts as sites of independent pleasure, sexual authority (Lee, 1997), joy (McCann, 2020), and positive femininity (Millsted & Frith, 2003). In the past several years, these perspectives have been taken up by

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<https://doi.org/10.1016/j.bodyim.2026.102054>

Received 12 December 2025; Received in revised form 16 February 2026; Accepted 16 February 2026

Available online 24 February 2026

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researchers with an interest in positive body image – an “overarching love and respect for the body” (Wood-Barcalow et al., 2010, p. 112) – resulting in the identification of the breast appreciation construct.

Breast appreciation is a form of positive breasted experience, which involves “accepting, holding favourable opinions toward, and respecting one’s breasts and their functions, while also rejecting media-promoted breast ideals as the only form of beauty” (Swami et al., 2022, p. 277). In this view, breast appreciation is a facet of the broader positive body image construct (see Tylka, 2018; & Tylka & Wood-Barcalow, 2015a), similar to analogous facets such as overall body appreciation (Tylka & Wood-Barcalow, 2015b) and functionality appreciation (Alleva et al., 2017). To measure the construct of breast appreciation, Swami et al. (2022) developed the 9-item Breast Appreciation Scale (BrAS). To do so, they initially developed a pool of 14 items tapping a putative breast appreciation construct. Based on responses to these items by cisgender women in the United Kingdom, which were subjected to an exploratory factor analysis (EFA) and scale purification methods, these authors retained a set of nine items.

The results of the EFA by Swami et al. (2022) indicated that the nine retained items formed a unidimensional factor with adequate composite reliability. This 9-item model of the BrAS was also cross-validated by Swami et al. (2022) using confirmatory factor analysis (CFA) in a new sample of cisgender women from the United Kingdom. Additionally, Swami et al. (2022) also reported that (manifest) scores on the BrAS has adequate convergent validity (i.e., positive, moderate associations with measures of positive body image and negative, moderate associations with breast size dissatisfaction and body dissatisfaction), concurrent validity (i.e., positive, moderate associations with self-esteem, self-compassion, mental health status, and breast awareness), discriminant validity (i.e., no significant association with impression management), and incremental validity (breast appreciation scores predicted unique variance in breast awareness above-and-beyond associations with breast size dissatisfaction).

To date, the BrAS has been validated in a small handful of national and linguistic groups beyond the United Kingdom. Thus, in their study Swami et al. (2022) also reported that the BrAS achieved full scalar invariance across samples of English-speaking, cisgender women from the United Kingdom, United States, and Australia. Beyond English-speaking samples, the BrAS has also been validated with women in Romania (Swami et al., 2024), Colombia (Mebarak et al., 2025), and Lebanon (Hallit et al., 2026). In each of these cases, researchers were able to retain a unidimensional model with all nine items based on the combined results of EFA and CFA (Hallit et al., 2026; Mebarak et al., 2025; Swami et al., 2024). These studies also reported that (latent) breast appreciation scores showed adequate evidence of convergent and concurrent validity. Thus, the available evidence suggests that the 9-item BrAS adequately captures the breast appreciation construct across some national and linguistic groups, with adequate evidence of reliability and validity in these communities of women.

At the same time, however, for the BrAS to be accepted as a valid and reliable tool across diverse communities, it is necessary to extend the number of communities in which the instrument’s psychometric properties have been assessed. This is particularly important because breasted experiences are likely shaped by a myriad of systems, values, and factors that vary across national and linguistic groups, such as cultural norms about the (symbolic) importance of breasts, meaning-making around femininity and sexuality, and the strength of patriarchal and heteronormative systems that limit ownership and representation of the breasts (Bartlett, 2000; Naugler, 2009; Swami et al., 2020). From this point-of-view, it should not be assumed that the psychometric indices of the BrAS will necessarily be replicated across national and linguistic groups. Rather, it remains important to consider the instrument’s psychometric properties in women from diverse cultural contexts.

1.1. The Canadian context

As a contribution to knowledge, the present study examined the psychometric properties of the BrAS in French-speaking Canadian women in Quebec, the largest province in Canada by area. Beyond simply extending the focus to include a new cultural and linguistic context, there are several reasons why focusing on French-Canadian women may be useful. First, as in most parts of North America, women’s breasts have been and still are normatively commodified and sexually objectified in Canada (Davey, 2022). At the same time, however, Canadian grassroots feminism has historically challenged patriarchal ownership over the breast (e.g., in emphasising breast self-examination as a tool of feminist resistance) and fostered liberationist approaches toward the female body in both social and medical spheres (Patton & Wood, 2022). In Quebec, in particular, Québécois feminist authors have – since at least the 1970s – shaped contemporary discourse around women’s bodies, patriarchal systems, and feminist alternatives to reclaiming the body (Andersen, 1988).

What this means in an everyday, practical sense is more difficult to discern, but it is likely that French-Canadian women’s breasted experiences are complex. For instance, it is apparent that performativities around breasted experiences are contested both at systemic (e.g., Morton, 2011) and everyday levels by Canadian women (Davey, 2022). At a systemic level, for instance, Arneil (2000) has discussed how Canadian legal frameworks have enshrined the breasts as non-sexual, multi-purpose organs, even if they remain sexualised in popular culture. At a more everyday level, Davey (2022) explored, through interviews with Canadian women, how lived experiences of being braless required negotiations of heteronormative expectations around the contained female body and vulnerabilities in the face of abjection. At the same time, however, going braless afforded Davey’s (2022) participants opportunities to prioritise their own physical comfort and thereby experience greater embodiment.

Beyond these issues, there is a further reason why availability of a French version of the BrAS may be useful. Breast cancer is the most common cancer and the second leading cause of mortality from cancer in Canadian women (Canadian Cancer Statistics Advisory Committee, 2023). Each of the Canadian provinces has adopted population-based mammography screening programmes, aimed at detecting breast cancer at an earlier stage and thereby reducing mortality rates. In Quebec, the programme – initiated in 1998 – is known as the *Programme québécois de dépistage du cancer du sein* (Quebec Breast Cancer Screening Program) and offers free mammography screening for all women aged 50–69 years. Although the programme aims to reach and maintain a 70% participation level, rates have rarely exceeded 50% annually (Institut national de santé publique du Québec, 2006). While there are likely complex reasons for this, availability of a validated measure of breast appreciation would allow scholars to examine the extent to which positive breasted experiences may (or may not) play a role in screening uptake decisions.

1.2. The present study

In the present study, we assessed the psychometric properties of a novel French translation of the BrAS in a sample of French-Canadian (Québécois) women. Following previous work (Hallit et al., 2026; Mebarak et al., 2025; Swami et al., 2022, 2024), we utilised an EFA-to-CFA analytic strategy, which involves a first-pass EFA step that is followed by a cross-validation using CFA. This analytic strategy is recommended for the test adaptation of instruments that are presumed to have a unidimensional factor structure (Swami & Barron, 2019). It allows for a test of factor structure without the imposition of theoretical and modelling restrictions (i.e., a data-driven approach to ascertaining factor structure using EFA) and a cross-validation where all indicators (items) are assumed to correspond to a single factor (i.e., a theory-driven approach to ascertaining factor structure using CFA; Swami, Maiano

et al., 2023). Here, we hypothesised that the BrAS would evidence a unidimensional factor structure with all nine of its items retained.

In addition, we also examined invariance of the optimal factor solution of the BrAS across two independent split-half subsamples and examined differential item functioning (DIF) of the BrAS as a function of women's age, as well as its invariance as function of subsamples. Tests of DIF provide an index of measurement invariance where characteristics are continuous or when multiple variables are assessed simultaneously (Penfield & Lam, 2000). Establishing DIF in relation to BrAS responses is important, firstly, because previous work has not considered the possibility that respondent age may impact the way in which BrAS items are understood and completed. Certainly, there is some evidence that there is a weak, positive relationship between breast appreciation and women's age (Mebarak et al., 2025; Swami et al., 2022). However, it is currently unknown whether respondent age differentially affects responding on BrAS items; this is something we hoped to rule out in the present study.

Finally, we also assessed convergent and concurrent validity of the French BrAS using additional instruments that have been previously validated for French-speaking populations. Further, many of these measures were included in the original BrAS original study, and they and others were chosen because of their established relations with breast-related body image, positive body image, other embodied experiences (i.e., eating behavior), and well-being and distress. More specifically, moderate associations between higher breast appreciation and greater body appreciation and lower breast size dissatisfaction, respectively, would provide evidence of convergent validity. In addition, moderate associations between breast appreciation and higher self-compassion, life satisfaction, "healthy" orthorexia, and self-esteem, as well as lower orthorexia nervosa, anxiety, depression, and dieting, respectively, would provide evidence of concurrent validity.

2. Method

2.1. Participants

The initial sample consisted of 408 French-speaking Canadian women from Quebec. However, we excluded women who had a diagnosis of an eating disorder (5.4%) and/or who had received such a diagnosis in the past (8.7%), as well as women who reported being pregnant at the time of survey completion (0.7%), given that disordered eating patterns or pregnancy may alter the size, shape, and women's experience of their breasts. The remaining sample, therefore, consisted of 365 women (aged between 17 and 90 years, $M = 51.46$, $SD = 15.98$). In terms of educational qualifications, 0.5% had completed elementary education, 13.5% had completed high school education, 15.9% had completed a diploma of vocational studies, 23.6% had completed a pre-university programme (CEGEP), 30.8% had an undergraduate degree, and 15.7% had a postgraduate degree. Finally, self-reported body mass index of participants ranged from 11.48 to 53.75 kg/m² ($M = 27.26$, $SD = 7.05$).

2.2. Measures

2.2.1. Participant information

Women were asked to self-report their age, educational qualifications, weight and height (used to compute body mass index), current or past diagnosis of eating disorder, and if they were pregnant. Because use of body mass index may perpetuate weight stigma (Bailey et al., 2025), we elected to report participant body mass index for descriptive purposes only and not to include these data in analyses.

2.2.2. Breast appreciation

Preparation of a French translation of the BrAS from its original English version was based on Swami and Barron's (2019) test adaptation recommendations. First, the instructions, response categories, and items

of the BrAS were separately translated from English to French by a professional translator and an author of the present study, who is bilingual. Second, a member of the research team examined the two French translations, resolved minor discrepancies between translations, and produced a synthesised French version. Third, two independent professional translators back-translated the synthesised version into English. Finally, a committee (i.e., comprising all translators and members of the research team) examined, discussed, and resolved minor discrepancies between the forward- and back-translations. The final version of the French BrAS is presented in the [Supplementary materials](#).

2.2.3. Breast size dissatisfaction

To assess women's breast size dissatisfaction (i.e., the discrepancy between actual and ideal breast size), we used the Breast Size Rating Scale (BSRS; Swami et al., 2015; French translation: Swami et al., 2020). The BSRS is a figural rating scale comprising 14 computer-generated images of women with increasing breast size. Women were asked to select the image on a 14-point scale (ranging from 1 = *smallest breast size* to 14 = *largest breast size*) that most closely matched their current and ideal (i.e., would most like to possess) breast size. A score of breast size dissatisfaction was calculated by taking the absolute difference between current and ideal breast size ratings. Higher scores reflect greater breast size dissatisfaction (Swami et al., 2015) and have been previously shown to have adequate convergent and concurrent validity (Swami et al., 2015), including in French-speaking adults (Swami et al., 2020). Because the BSRS comprises only two ratings, it is not possible to compute composite reliability for this instrument.

2.2.4. Body appreciation

The Body Appreciation Scale-2 (BAS-2; Tylka & Wood-Barcalow, 2015; French translation: Kertechian & Swami, 2017) was used to assess women's acceptance of the body, respect and care for the body, and protection of the body from unrealistic beauty standards. Women were asked to rate the 10 items of the BAS-2 on a 5-point scale (1 = *never*, 5 = *always*), with higher scores reflecting greater body appreciation. The BAS-2 has been shown to have acceptable psychometric properties (i.e., a unidimensional factor structure, adequate composite reliability, and good construct validity) in French-speaking Canadian adults (Swami, Tran, et al., 2023). The composite reliability coefficient (McDonald's ω) of scores on this scale was .965 in the present study.

2.2.5. Dieting

Women's dieting-related attitudes and behaviours characteristic of disordered eating were assessed using the Dieting subscale of the Eating Attitudes Test-26 (EAT-26; Garner et al., 1982; French translation: Lechner et al., 1994). Women were asked to rate the 13 items of the Dieting subscale on a 6-point scale (1 = *never*, 6 = *always*), with higher scores reflecting greater dieting-related attitudes and behaviours. Scores on the EAT-26 (including the Dieting subscale) have demonstrated adequate psychometric properties (i.e., construct validity and composite reliability) in French-speaking Canadian adults (Lechner et al., 1994). The composite reliability coefficient (McDonald's ω) of scores on this subscale was .926 in the present study.

2.2.6. Orthorexia

The Teruel Orthorexia Scale (TOS; Barrada & Roncero, 2018; French translation: Maïano et al., 2022) was used to assess women's symptoms of orthorexia nervosa and "healthy" orthorexia. This 17-item instrument is based on Barrada and Roncero's (2018) model of orthorexia, which distinguishes between an extreme preoccupation with eating healthily (i.e., orthorexia nervosa; 8 items) and a healthy interest in diet, self-assessed healthy behaviours with regard to diet, and eating "healthily" as part of one's core identity (i.e., non-psychopathological or "healthy" orthorexia"; 9 items). Women rated items on a 4-point scale (0 = *completely disagree*, 3 = *completely agree*), with higher scores reflective of greater orthorexic symptomatology. Scores on the TOS have

been shown to have a 2-factor structure and adequate psychometric properties in French-speaking Canadian adults (Maïano et al., 2022). The composite reliability coefficients (McDonald's ω) of scores on the Orthorexia Nervosa and Healthy Orthorexia subscales were .858 and .872 in the present study, respectively.

2.2.7. Anxiety and depression

Women's symptoms of anxiety (7 items) and depression (7 items) were measured using the 14-item Hospital Anxiety and Depression Scale (HADS; Zigmond & Snaith, 1983; French translation: Savard et al., 1998). Participants rated their symptoms on a 4-point scale, with total scores for each subscale ranging from 0 to 21. Scores on the HADS have been shown to have good sensitivity in French-speaking Canadian adults to determine the presence of anxiety or depressive symptoms, respectively (Savard et al., 1998). Composite reliability coefficients (McDonald's ω) on the Anxiety and Depression subscales were .832 and .820 in the present study, respectively.

2.2.8. Life satisfaction

Women's subjective well-being was assessed using the 5-item Satisfaction With Life Scale (SWLS; Diener et al., 1985; French translation: Blais et al., 1989). The SWLS items were rated on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*), with higher scores reflecting greater life satisfaction. The SWLS has been shown to have adequate psychometric properties (i.e., unidimensional factor structure, adequate composite reliability, and good construct validity) in French-speaking Canadian adults (Swami, Stieger et al., 2025). The composite reliability coefficient (McDonald's ω) for SWLS scores was .913 in the present study.

2.2.9. Self-compassion

Self-compassion was measured using the Short Form of the Self-Compassion Scale (SCS-SF; Raes et al., 2011; French translation: Swami, Tran et al., 2025). Women rated the 12-item SCS-SF using a 5-point scale (1 = *almost never*, 5 = *almost always*). In their recent work, however, Swami, Tran et al., (2025) suggested that, in diverse populations including French-speaking Canadian adults, the SCS-SF is best conceptualised as consisting of two discrete subscales assessing compassionate (5 items) and uncompassionate self-responding (6 items). As such, only the compassionate self-responding subscale was used in the present study and the composite reliability coefficient (McDonald's ω) for scores on this subscale was .796.

2.2.10. Self-esteem

Women complete the Single-Item Self-Esteem Scale (SISE; Robins et al., 2001; French translation: Swami et al., 2020). They rated a single item ("I have high self-esteem") on a 7-point scale (1 = *strongly disagree* to 5 = *strongly agree*), with higher scores reflecting higher self-esteem. Evidence of construct validity of the SISE has been reported in diverse populations including French-speaking Canadian adults (Swami et al., 2020).

2.3. Procedure

The research ethics committee of the first and last authors' university (approval code: #2025–3656) authorised this study. The present study was conducted between November 2024 and April 2025 following the principles of the Declaration of Helsinki. Women were primarily recruited in the Canadian province of Quebec via a panel from Leger Opinion (95% of the total sample). For their participation they received CAD\$1.80 (95% of the total sample). An additional 5% of the total sample, who did not receive any compensation, was recruited in the community via advertising placed on social media, a local newspaper, and the websites of private clinics. The survey was hosted on LimeSurvey and all women completed the survey voluntarily after providing digital informed consent. The order of measures in the survey was pre-

determined and in the order described above.

2.4. Analytic strategy

The total sample was randomly split into two separate subsamples (182 and 183) to facilitate an EFA-to-CFA strategy. No significant differences were found between the two split-half subsamples as function of age, $t(363) = 1.54$, $p = .124$, $d = .37$, and educational qualifications, $\chi^2(9) = 6.17$, $p = .723$.

2.4.1. First split-half subsample

All analyses were performed using Mplus 9 (Muthén & Muthén, 2025). As recommended by Hair et al. (2009), data factorability was assessed using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (which should ideally be $\geq .80$) and Bartlett's test of sphericity (which should be significant). EFAs with one to three factors were performed using the full information maximum likelihood robust (MLR) estimator (missing data at the BrAS item level ranged from 0% to 7.14%, $M = 2.14\%$), an oblique geomin rotation, and an epsilon value of .5 (Marsh et al., 2009, 2014). Horn's (1965) parallel analysis test was used to assist factor retention decisions and was complemented as recommended (Hu & Bentler, 1999; Marsh et al., 2005) using the Comparative Fit Index (CFI; .90 for acceptable fit to the data and .95 for good fit to the data, respectively), the Tucker Lewis Index (TLI; same thresholds as the CFI), and the root mean square error of approximation (RMSEA; .08 for acceptable and .06 for good fit to the data, respectively). Finally, McDonald's (1970) ω was used to estimate composite reliability of BrAS latent factor(s).

2.4.2. Second split-half subsample

A CFA was performed with Mplus 9 (Muthén & Muthén, 2025) and using the MLR estimator (missing data at the BrAS item level ranged from 0% to 2.19%, $M = 1.00\%$). In this CFA model the number of latent factors to be estimated was based on the best EFA solution retained with the first split-half subsample. The same fit indices as those used with the first split-half subsample was used to assess model fit.

2.4.3. First and second split-half subsamples

Measurement invariance of the optimal CFA model across split-half subsamples was tested using Mplus 9 (Muthén & Muthén, 2025) and the MLR estimator. As recommended by Millsap (2011), we tested for: (a) configural invariance, and invariance of (b) loadings, (c) intercepts, (d) uniqueness, (e) latent variance, and; (f) latent mean(s). Each model fit was compared to the previous one based on changes (Δ) on CFI, TLI, and RMSEA. Δ CFI and/or Δ TLI decreases lower than .01, and/or an increase in Δ RMSEA lower than .015, provides support for invariance (Chen, 2007; Cheung & Rensvold, 2002).

The presence of DIF and of latent mean differences as function of age (standardised prior the analyses), as well as the invariance of this predictor across split-half subsamples, was examined using a hybrid MIMIC multiple-group model (Maïano et al., 2023; Morin et al., 2018). This method was retained as it is recognized as the most optimal when the variable of interest is continuous such as age. The hybrid MIMIC model was developed based on the most invariant model identified during the measurement invariance test of the BrAS across split-half subsamples. As recommended (Morin, 2023; Maïano et al., 2023), the following hybrid MIMIC models were tested: (a) null effects, (b) saturated, and (c) factor only. These models were first examined with all predictions freely estimated across split-half subsamples, and then, in the most appropriate hybrid MIMIC model, they were constrained to be equal across split-half subsamples.

An increase of Δ CFI and/or Δ TLI $\geq .01$ and/or a decrease of Δ RMSEA $\geq .015$ for the saturated model relative to the null effects model suggests that there is a meaningful association between age and item responses on the BrAS. When this is the case, similar fit statistics between the saturated and the factor only models suggests that the

predictions are limited to the factor. However, an improvement of goodness of fit statistics for the saturated model relative to the factor only model suggests the presence of DIF. Finally, the associations between item responses on the BrAS and age are invariant across split-half subsamples when we observe a lack of substantial improvement between the most appropriate hybrid MIMIC model and an alternative version of this model constraining the examined predictions to be equal across split-half subsamples.

Convergent and concurrent validity of the BrAS was examined in the total sample using a structural equation model (SEM). In this model, the BrAS latent factor(s) was estimated based on the best CFA solution and was correlated with the observed scores of measures of breast size dissatisfaction, body appreciation, dieting-related attitudes and behaviours, orthorexia, symptoms of anxiety and depression, life satisfaction, self-compassion, and self-esteem. Additionally, in this SEM, all observed scores were correlated altogether.

3. Results

3.1. First split-half subsample

Both Bartlett's test of sphericity, $\chi^2(36) = 2755.73, p < .001$, and the Kaiser-Meyer-Olkin measure of sampling adequacy (.939), indicated that the BrAS items had adequate common variance for factor analysis with the data from the first split-half subsample. The results of the EFAs are presented in Table 1. The unidimensional factor solution (Model 1-1) had acceptable-to-good-fit to the data based on CFI and TLI, but not RMSEA. The 2- and 3-factor solutions had acceptable-to-good fit to the data (Models 1-2 and 1-3). Additionally, as illustrated in Fig. S1 in the Supplementary materials, the parallel analysis test provided support for a unidimensional solution. Inspection of the 2-factor solution showed that the first factor comprised five items and the second factor comprised four items, one of which had an improper standardised factor loading. Finally, inspection of the 3-factor solution showed that the first factor comprised two items, the second factor four items, and the third factor three items, of which two had loadings $< .40$.

Therefore, the unidimensional model was retained based on the analyses of fit indices, comparison with the other factor solutions, and the results of the parallel analysis test. As illustrated in Table 2, the standardised factor loading values were high ($\lambda = .718-.916$; $M_\lambda = .830$) and the composite reliability coefficient of the BrAS latent factor was excellent ($\omega = .953$). Nevertheless, parameters (i.e., errors of measurement, standardized residuals for covariances, modification indices, etc.) of the model were more carefully inspected to determine sources of misspecification of the unidimensional model. This revealed that the misspecification of the model could be explained by Items 1, 3, and 5 (i.e., higher measurement error values, higher values of standardised residuals for covariances, and higher modification indices for correlated uniqueness). Therefore, an EFA model with one to two factors excluding these items (#1, 3, and 5) was rerun (Model 2-1), with the results providing good-fit to the data for solutions with one to three factors (see Table 1). The parallel analysis test from this new EFA provided support for the unidimensional solution. The standardised factor loading values of this 6-item version of the BrAS remained high ($\lambda = .792-.942$; $M_\lambda = .851$) and the composite reliability coefficient of the BrAS latent factor remained excellent ($\omega = .941$).

3.2. Second split-half subsample

To rule out the possibility that the lack-of-fit of the original 9-item unidimensional model of the BrAS (Model 3-1) was spurious, we tested the fit of this model using CFA with the second split-half subsample (see Table 1). The results showed that this model had poor fit to the data for TLI and RMSEA. The standardised parameters (i.e., loadings and uniqueness) and composite reliability coefficient of this 9-item model are presented in Table 2. As in the first split-half subsample,

Table 1
Goodness-of-fit statistics for the BrAS.

Models	Samples	N°	Description	R^2	df	CFI	TLI	RMSEA	RMSEA 90 % CI		CM	ΔR^2	df	p	ΔCFI	ΔTLI	$\Delta RMSEA$
									LB	UB							
EFA	First-split half	1-1	1 factor - 9 items	76.256*	27	.950	.933	.100	.074	.127	-	-	-	-	-	-	-
		1-2	2 factors - 9 items	37.511*	19	.981	.964	.073	.037	.107	-	-	-	-	-	-	-
		1-3	3 factors - 9 items	21.577*	12	.990	.971	.066	.012	.110	-	-	-	-	-	-	-
CFA	Second-split half	2-1	1 factor - 6 items	18.758*	9	.983	.971	.077	.025	.127	-	-	-	-	-	-	-
		2-2	2 factors - 6 items	5.427	4	.997	.990	.044	.000	.0127	-	-	-	-	-	-	-
		3-1	1 factor - 9 items	114.861*	27	.919	.893	.133	.109	.159	-	-	-	-	-	-	-
		3-2	1 factor - 6 items	14.819	9	.991	.985	.059	.000	.112	-	-	-	-	-	-	-
		4-1	Configural invariance	33.452*	18	.987	.979	.069	.030	.104	-	-	-	-	-	-	-
Measurement invariance across samples	First and second-split half	4-2	Loadings invariance	42.634*	23	.984	.979	.068	.035	.100	4-1	9.16	5	.10	-.003	.000	-.001
		4-3	Intercepts invariance	45.677*	28	.985	.984	.059	.024	.089	4-2	2.02	5	.85	+.001	+.005	-.009
		4-4	Uniqueness invariance	51.847*	34	.985	.987	.054	.019	.082	4-3	7.06	6	.31	.000	+.003	-.005
		4-5	Variance invariance	52.250*	35	.986	.988	.052	.016	.080	4-4	.04	1	.83	+.001	+.001	-.002
		4-6	Latent mean invariance	52.527*	36	.986	.989	.050	.013	.078	4-5	.01	1	.96	.000	+.001	-.002
DIF: Age	First and second split half	5-1	Null effects	62.660	48	.989	.990	.041	.000	.067	-	-	-	-	-	-	-
		5-2	Saturated	52.931*	36	.987	.985	.051	.014	.078	5-1	8.74	12	.73	-.002	-.005	+.010
		5-3	Factor only	61.600	46	.988	.989	.043	.000	.069	5-2	7.91	10	.64	+.001	+.004	-.008

Note. BrAS = Breast Appreciation Scale; R^2 = robust maximum likelihood chi-square; df = degrees of freedom; CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; 90 % CI = 90 % confidence interval of the RMSEA; LB = lower bound; UB = upper bound; Δ = change from the previous model; ΔR^2 = robust chi-square difference tests (calculated from loglikelihoods for greater precision); EFA = exploratory factor analyses; CFA = confirmatory factor analyses; DIF = differential item functioning.
* $p \leq .05$.

Table 2

Standardised parameters estimates from the exploratory and confirmatory factor analytic representations of the BrAS in the first and second split half subsamples.

Items	First-split half				Second-split half			
	EFA - 9 items		CFA - 6 items		CFA - 9 items		CFA - 6 items	
	λ	δ	λ	δ	λ	δ	λ	δ
1	.852	.274	-	-	.801	.358	-	-
2	.912	.168	.912	.169	.846	.284	.832	.308
3	.718	.485	-	-	.762	.419	-	-
4	.790	.375	.796	.366	.817	.333	.815	.335
5	.781	.391	-	-	.742	.449	-	-
6	.858	.263	.831	.310	.797	.365	.780	.392
7	.808	.348	.792	.373	.854	.271	.840	.294
8	.916	.161	.942	.113	.915	.162	.934	.127
9	.832	.307	.832	.308	.867	.248	.880	.225
ω	.953		.941		.950		.939	

Note. BrAS = Breast Appreciation Scale; EFA = exploratory factor analyses; CFA = confirmatory factor analysis; λ = factor loadings; δ = uniquenesses; ω = McDonald's omega.

misspecification of the model could be explained by Items #1, 3, and 5. Therefore, these items were excluded and the CFA model was rerun. Results presented in Table 1 showed an acceptable fit to the data. As illustrated in Table 2, results showed that factor loading values were high ($\lambda = .780-.934$; $M_{\lambda} = .847$) and the composite reliability coefficient of the BrAS latent factor was excellent ($\omega = .939$).

3.3. First and second split-half subsamples

3.3.1. Subsample invariance

Goodness-of-fit statistics from tests of measurement invariance of the 6-item version of the BrAS are presented in Table 1 (Models 4-1 to 4-6). Results provided support for complete measurement invariance (i.e., configural, loadings, intercepts, uniqueness, variance and latent mean) of the 6-item unidimensional CFA model of the BrAS.

3.3.2. Differential item functioning across split-half subsamples

The goodness-of-fit statistics of the hybrid MIMIC models are provided in Table 1. These analyses were built from the most invariant model (across split-half subsamples) identified in previous analyses (i.e., Model 4-6: latent invariant model). The current set of results first showed that the saturated (Model 5-2) and factor only (Model 5-3) models do not have a better fit to the data than the null effects model (Model 5-1), suggesting a lack of association between age and BrAS item responses. Additionally, the fit of the factor only model (Model 5-3) was comparable ($\Delta CFI < .01$; $\Delta TLI < .01$; $\Delta RMSEA < .015$) to that of the saturated model (Model 5-2), suggesting a lack of DIF. Finally, the invariant model was not tested as no association was found between age and the BrAS item responses.

3.3.3. Convergent and concurrent validity

The SEM examining the convergent and concurrent validity of the 6-item unidimensional model representation of the BrAS provided a good level of fit to the data: $\chi^2(59) = 66.004$, CFI = .997, TLI = .995, RMSEA = .018 (90 % CI = .000; .038). As illustrated in Table 3, results revealed significant and positive relationships between breast appreciation and body appreciation, self-compassion, life satisfaction, healthy orthorexia, and self-esteem. Conversely, significant and negative relationships were found between breast appreciation and breast size dissatisfaction, anxiety, depression, and dieting, respectively. No significant association was found with orthorexia nervosa.

4. Discussion

In the present study, we sought to validate a novel French translation of the 9-item BrAS (Swami et al., 2022) in a sample of French-Canadian women. First, in terms of its factor structure, our results were somewhat discrepant from previous test adaptation studies of the BrAS (Hallit

Table 3

Construct validity of the BrAS in the overall sample.

	BrAS
Breast size dissatisfaction	-.398*
Body appreciation	.497*
Self-compassion	.305*
Life satisfaction	.304*
Healthy orthorexia	.225*
Orthorexia nervosa	-.073
Anxiety	-.277*
Depression	-.287*
Self-esteem	.388*
Dieting	-.286*

Note. BrAS = Breast Appreciation Scale.

* $p \leq .01$.

et al., 2026; Mebarak et al., 2025; Swami et al., 2024), insofar as we were only able to retain a unidimensional model with six of the nine BrAS items. Nevertheless, this model was invariant across both split-half subsamples, suggesting a degree of cross-sample stability, and showed no signs of DIF as function of age. Furthermore, latent scores on the 6-item unidimensional model of the BrAS showed strong evidence of convergent and concurrent validity, given that correlations with additional variables included in the survey package were in the expected directions and strength ranges. We consider these results in further detail below, beginning with the discrepant factor analytic results.

Previous studies examining the factor structure of the BrAS have supported the original study (Swami et al., 2022) in retaining a unidimensional model with all nine items (Hallit et al., 2026; Mebarak et al., 2025; Swami et al., 2024). However, the results of our first-pass EFA indicated that a unidimensional problem had somewhat less-than-adequate fit (specifically, RMSEA above an acceptable cut-off). On the other hand, 2- and 3-factor models had good fit to the data, but evidenced problematic factor solutions, including improper standardised factor loadings (the 2-factor solution) or unstable factors with weak item-factor loadings (the 3-factor solution). In view of the results of the parallel analysis test pointing to the retention of a single factor, we examined parameters of the unidimensional model more closely, which indicated that model misspecification could be attributed to three items (Items #1, 3, and 5). Exclusion of these three items from a subsequent EFA indicated that a unidimensional model achieved good fit to the data.

Additionally, our results also showed that the original 9-item model of the BrAS had poor CFA-based fit in the second split-half subsample. It is possible that fit could be improved by relaxing constraints for certain item pairs, but such *ex-post-facto* model modifications (i.e., the addition of correlated uniquenesses; CUs) should be avoided (Morin, 2023; Schweizer, 2012). Indeed, CUs should be implemented in an *a priori*

manner (Morin, 2023; Swami, Maïano, et al., 2023) when surveys are comprised of a mixture of positively and negatively worded items or when surveys use parallel wording. This was not the case with the items that were identified as problematic in the present work (i.e., Items #1, 3, and 5) and, as such, we elected not to add *post hoc* modifications to the model, which would reduce its replicability and generalisability.

Conversely, when we examined the CFA-based fit of the 6-item model in the second split-half subsample, we found that the model had adequate fit to the data and, additionally, was fully invariant across both split-half subsamples. Overall, then, the present results indicate that, in our sample of French-Canadian women, a model that retains six of the nine BrAS items shows adequate factorial validity. It is difficult to know why the three items functioned poorly in the present sample. Given that previous studies have reported no similar issues (Hallit et al., 2026; Mebarak et al., 2025; Swami et al., 2022, 2024), one preliminary conclusion is that this discrepancy is reflection of linguistic issues, such as the way that gratitude or self-love manifests. For example, it may be less appropriate or easy in French to express love or appreciation for one's breasts. Another possibility is that the content of these three items, although somewhat varied, are less relevant for French-Canadian women, which resulted in poorer item functioning.

From a practical point-of-view, these results are not necessarily problematic, as our results suggest that the unidimensional, 6-item model was stable across our split-half subsamples. Of course, difficulties will arise should researchers wish to compare latent breast appreciation across existing studies or conduct future cross-national comparisons, where measurement invariance may not be achievable. One interim solution would be to conduct a qualitative investigation of understandings and meanings of breast appreciation in French Canadian women. In a similar vein, it may be useful to ask participants to discuss specific BrAS items to determine whether understandings of items vary across linguistic or national groups. Such work may help scholars better understand the extent to which the present results reflect underlying cultural or linguistic idiosyncracies in the way that breast appreciation, as measured using the BrAS, is experienced.

Aside from factorial validity, the present study also examined – for the first time – DIF of the BrAS as a function of participant age. Our analyses indicated a lack of DIF as a function of age, a result that was invariant across both split-half subsamples. We also found no significant association between breast appreciation and age, which stands in contrast to previous work indicating that breast appreciation is weakly and positively associated with age (Mebarak et al., 2025; Swami et al., 2024). Importantly, given that previous studies have not controlled for measurement error in correlational analyses, it is likely that associations were inflated and closer to negligible effects in real-world terms. As with the issues around factorial validity, qualitative research may be better suited to understanding the extent to which breast appreciation changes across the lifespan.

Finally, and consistent with previous work (Hallit et al., 2026; Mebarak et al., 2025; Swami et al., 2022, 2024), our results indicated that scores on the French BrAS exhibited adequate convergent and concurrent validity. More specifically, breast appreciation was moderately and positively associated with body appreciation, which is consistent with the view that these are related facets of positive body image (Swami et al., 2022). Additionally, breast appreciation was moderately and positively associated with indices of psychological well-being (i.e., life satisfaction, self-esteem) and negatively associated with breast size dissatisfaction and indices of ill-being (i.e., depression, anxiety). Finally, we also found that greater breast appreciation was moderately associated with eating-related attitudes and behaviours (negatively with dieting and positively with healthy orthorexia). In broad outline, these results support the nomological validity of the French BrAS.

There were several limitations of the present work that should be considered. First, our assessment of the psychometric properties of the French BrAS could be extended in several ways. For example, despite

that the sample size was acceptable regarding the complexity of the tested model, it remains near the minimal. Therefore, the present results need to be replicate in a larger sample of French-Canadian Women. Additionally, we did not examine measurement invariance across motherhood status, which may provide an index of known-groups validity (Swami et al., 2022), and we also did not assess indices of discriminant and predictive validity. In a similar vein, we did not examine respondents' scores across time, which means we are unable to provide an assessment of temporal stability or test-retest reliability. Second, in our recruitment method we used an online panel and online community sampling which means that we did not recruit a representative sample of Québécois women. Our sampling strategy may have also inadvertently introduced sampling biases (e.g., some participants being more likely to respond to an advertisement about a survey on body image), which again limits the generalisability of our findings. In a similar vein, greater confidence in the validity of our data would have been provided through the use of attention checks in the survey and randomisation of the order of presentation of instruments to minimise potential order effects.

Other limitations of our study include the lack of comprehensive data that were gathered on respondent sociodemographic characteristics, such as information about urbanicity, socioeconomic status and food insecurity, and religion or religiosity, which may affect the ways in which breast appreciation is understood and experienced. We also recognise that our sample was restricted to French-Canadian women, and, in future work, it will be important to extend our findings to other communities in Quebec, such as First Nation peoples. In future work, it may also be useful to adopt more emic approaches (see Swami & Barron, 2019), which would allow researchers to better understand whether French-Canadian communities understand or experience breast appreciation in unique ways. As we have described above, this may go some way to helping scholars understand the issue around some poorly functioning items in this context.

These limitations aside, the present study suggests that it is possible to derive a unidimensional model of the BrAS for use in French-Canadian women. This unidimensional model retains six of the nine original BrAS items and demonstrates adequate composite reliability, a lack of DIF as a function of age, and adequate convergent and concurrent validity. Overall, then, we conclude that the French BrAS can be used with confidence for the assessment of breast appreciation in French-Canadian women, though scholars should be cognisant of issues around item functioning in this context. As research on breast appreciation continues to grow, fuller research of the ways in which understandings of this construct varies – or does not vary – across cultural and linguistic communities will become increasingly important.

CRedit authorship contribution statement

Christophe Maïano: Conceptualization, Methodology, Validation, Formal analysis, Data Curation, Resources, Writing – original draft, Writing – review & editing, Software, Supervision, Visualization, Project administration, Investigation, Funding acquisition. **Viren Swami:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision. **Tracy L. Tylka:** Conceptualization, Methodology, Validation, Writing – review & editing. **Annie Aimé:** Conceptualization, Methodology, Validation, Writing – review & editing.

Ethics Statement

This study was approved by the research ethics committee of the Université du Québec en Outaouais (#2025–3656) and conducted in accordance with the Declaration of Helsinki. Participants provided digital consent prior to responding to the online survey.

Funding

Payments to participants was by a research institutional fund awarded to Christophe Maïano. The funder has no role in the conceptualization, design, analysis, or writing of this article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. One of the authors of this article, Tracy L. Tylka, is the Editor-in-Chief of *Body Image*. She had no editorial involvement in the assessment, handling, or peer review of this article. Her access to this submission was blocked in her role as editor. Full responsibility for the editorial process for this article was delegated to another editor.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.bodyim.2026.102054](https://doi.org/10.1016/j.bodyim.2026.102054).

Data availability

The authors do not have permission to share data.

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